

Endpoint failure when cotype is not attained

Run fa_banach_001, agent_lane_15

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Abstract

Aron et al. prove an optimal Hardy–Littlewood theorem for range spaces that attain their cotype, and an ε -shifted theorem for finite-cotype spaces when cotype attainment is not known. We show that the epsilon loss is necessary in general. There is an explicit infinite-dimensional Banach space Y with $\cot Y = 2$, not of cotype 2, and a bounded linear operator $A : \ell_4 \rightarrow Y$ for which $(\|Ae_j\|)_j \notin \ell_4$.

1 Source context

The source paper is arXiv:1602.00178v4, “Optimal exponents for Hardy–Littlewood inequalities for m -linear operators” [1]. Its main vector-valued result assumes that the target Banach space Y attains its cotype.

ABSTRACT. The Hardy–Littlewood inequalities on ℓ_p spaces provide optimal exponents for some classes of inequalities for bilinear forms on ℓ_p spaces. In this paper we investigate in detail the exponents involved in Hardy–Littlewood type inequalities and provide several optimal results that were not achieved by the previous approaches. Our first main result asserts that for $q_1, \dots, q_m > 0$ and an infinite-dimensional Banach space Y attaining its cotype $\cot Y$, if

$$\frac{1}{p_1} + \dots + \frac{1}{p_m} < \frac{1}{\cot Y},$$

then the following assertions are equivalent:

- (a) There is a constant $C_{p_1, \dots, p_m}^Y \geq 1$ such that

$$\left(\sum_{j=1}^m \left\| \sum_{i=1}^{\infty} x_{ij} e_i \right\|_{p_i}^{q_i} \right)^{\frac{1}{q_1}} \leq C_{p_1, \dots, p_m}^Y \left\| \sum_{i=1}^{\infty} x_i e_i \right\|_Y,$$

Later, the paper treats the case where one does not know if Y attains the infimum of its cotypes, but the conclusion is weakened by replacing each endpoint exponent q_j by $q_j + \varepsilon$.

In the case that we do not know if Y attains the infimum of its cotypes, using the previous arguments, it is possible to prove the following:

Theorem 2.4. *Let $q_1, \dots, q_m > 0$ and Y be an infinite-dimensional Banach space with finite cotype. If*

$$\frac{1}{p_1} + \dots + \frac{1}{p_m} < \frac{1}{\cot Y},$$

then the following assertions are equivalent:

- (a) *There is a constant $C_{n, \dots, n}^{Y, \varepsilon} > 1$ such that*

The construction below shows that this endpoint loss is not merely an artifact of the proof.

2 Idea of the proof

If a Banach space has cotype s for every $s > 2$, but not cotype 2, then $\cot Y = 2$ is not attained. We build such a space from finite blocks $\ell_{2+1/n}^{N_n}$, choosing the dimensions N_n so large that the cotype 2 constants of the blocks diverge. The same oversized blocks let us construct a bounded block-diagonal operator $A : \ell_4 \rightarrow Y$. Each block is scaled just enough to keep the operator bounded, but not enough to make the endpoint coefficient sequence $(\|Ae_j\|)$ belong to ℓ_4 .

3 The counterexample

Theorem 1. *There is an infinite-dimensional Banach space Y with finite cotype and $\cot Y = 2$, but not of cotype 2, such that the endpoint linear Hardy–Littlewood inequality predicted by the epsilon-free version of Theorem 2.4 in [1] fails. More precisely, there is a bounded operator*

$$A : \ell_4 \longrightarrow Y$$

such that

$$\sum_{j=1}^{\infty} \|Ae_j\|_Y^4 = \infty.$$

Proof. For $n \geq 1$, put

$$r_n = 2 + \frac{1}{n}, \quad N_n = n^{4n+2}, \quad E_n = \ell_{r_n}^{N_n},$$

and define

$$Y = \left(\bigoplus_{n=1}^{\infty} E_n \right)_2.$$

First we record the cotype of Y . The space Y has cotype s for every $s > 2$. Indeed, for all sufficiently large n one has $r_n \leq s$, and the standard cotype theorem for L_p -spaces gives uniformly bounded cotype s constants for the blocks $\ell_{r_n}^{N_n}$ with $r_n \in [2, s]$; the finitely many remaining blocks are harmless. An ℓ_2 -sum of spaces with uniformly bounded cotype s again has cotype s . Hence $\cot Y \leq 2$, and since cotype exponents are always at least 2, we have $\cot Y = 2$.

But Y is not of cotype 2. Each block E_n is 1-complemented in Y . Applying a cotype 2 inequality to the unit vector basis of $\ell_{r_n}^{N_n}$ would force its cotype 2 constant to be at least

$$\frac{N_n^{1/2}}{N_n^{1/r_n}} = N_n^{1/2-1/r_n} = (n^{4n+2})^{1/(4n+2)} = n.$$

These lower bounds are unbounded, so Y cannot have cotype 2.

Now split the canonical basis of ℓ_4 into consecutive blocks of cardinalities N_n , and write $x^{(n)} \in \ell_4^{N_n}$ for the n -th block of $x \in \ell_4$. Define $A : \ell_4 \rightarrow Y$ blockwise by

$$(Ax)_n = \frac{1}{n} N_n^{-(1/r_n-1/4)} x^{(n)} \in \ell_{r_n}^{N_n},$$

where $x^{(n)}$ is identified with the same coordinate vector in $\ell_{r_n}^{N_n}$. Since the identity $\ell_4^{N_n} \rightarrow \ell_{r_n}^{N_n}$ has norm $N_n^{1/r_n-1/4}$, each block operator has norm at most $1/n$. Therefore

$$\|Ax\|_Y^2 \leq \sum_{n=1}^{\infty} \frac{1}{n^2} \|x^{(n)}\|_4^2 \leq \left(\sum_{n=1}^{\infty} \frac{1}{n^4} \right)^{1/2} \left(\sum_{n=1}^{\infty} \|x^{(n)}\|_4^4 \right)^{1/2},$$

so A is bounded from ℓ_4 to Y .

For a basis vector e_j in the n -th block,

$$\|Ae_j\|_Y = \frac{1}{n} N_n^{-(1/r_n - 1/4)}.$$

Thus

$$\sum_j \|Ae_j\|_Y^4 = \sum_{n=1}^{\infty} N_n \frac{1}{n^4} N_n^{-4(1/r_n - 1/4)} = \sum_{n=1}^{\infty} \frac{1}{n^4} N_n^{1-4(1/r_n - 1/4)}.$$

Since

$$1 - 4 \left(\frac{1}{r_n} - \frac{1}{4} \right) = 2 - \frac{4}{2 + 1/n} = \frac{2}{2n + 1},$$

we get

$$\frac{1}{n^4} N_n^{1-4(1/r_n - 1/4)} = \frac{1}{n^4} (n^{4n+2})^{2/(2n+1)} = 1.$$

Consequently $\sum_j \|Ae_j\|_Y^4 = \infty$.

For $m = 1$, $p = 4$, and $\cot Y = 2$, the endpoint exponent in the source notation is

$$\lambda_{1,2}^4 = \frac{2}{1 - \frac{2}{4}} = 4.$$

So the endpoint, epsilon-free version of Theorem 2.4 would assert precisely that $(\|Ae_j\|)_j \in \ell_4$ for every bounded $A : \ell_4 \rightarrow Y$, which the operator above disproves. $\square$

4 Verification notes

The included script `code/nonattained_cotype_endpoint_check.py` checks the exponent identities

$$N_n^{1/2 - 1/r_n} = n, \quad n^{-4} N_n^{1-4(1/r_n - 1/4)} = 1.$$

These identities are the two quantitative cores of the proof: the first shows non-attainment of cotype 2, and the second shows divergence of the endpoint ℓ_4 coefficient sum.

5 Novelty and limitations

Local run indexes were searched for arXiv:1602.00178, Hardy–Littlewood, optimal exponents, multilinear forms, cotype, and non-attained cotype. No exact prior packet was found. Bounded web searches on June 28, 2026 for the source arXiv id, Hardy–Littlewood cotype endpoint, and non-attained-cotype epsilon language found no separate known answer beyond the source arXiv record.

The packet does not contradict Theorem 2.4 of [1], because that theorem includes an ε -shift in every exponent. The result shows that this shift cannot be removed for arbitrary finite-cotype target spaces. The counterexample already occurs in the linear case $m = 1$; it therefore also blocks any multilinear theorem that would include the linear endpoint as a special case.

6 Human-review recommendation

The key review point is the standard cotype calculation for ℓ_2 -sums of finite-dimensional ℓ_r blocks. Once that is accepted, the endpoint failure is an explicit block-diagonal operator calculation.

References

- [1] R. M. Aron, D. Nunez-Alarcon, D. M. Pellegrino, and D. M. Serrano-Rodriguez, *Optimal exponents for Hardy–Littlewood inequalities for m -linear operators*, arXiv:1602.00178v4, 2016. <https://arxiv.org/abs/1602.00178>.